Smart Fencing System for Farm Security Using IoT

Objective

To develop an IoT-enabled smart fencing system that detects intruders around a farm, provides real-time alerts to the farmer, and enables remote monitoring and control through a mobile application.

Components Used

- ESP32 Microcontroller
- ESP32-CAM Module
- Ultrasonic Sensors
- Servo Motor
- Buzzer
- LED Indicators
- Smart Fencing Circuit (Non-Lethal Electric Deterrent)
- Blynk IoT Platform
- Wi-Fi Communication Module

Working Principle

Ultrasonic sensors installed along the farm boundary continuously monitor for intruders. When an object is detected within the predefined range, the sensor sends a signal to the ESP32 controller.

The ESP32 immediately activates the smart fencing circuit, buzzer, and LED indicators to deter the intruder. Simultaneously, a notification is sent to the farmer through the Blynk mobile application.

An ESP32-CAM mounted on a continuously rotating servo motor provides visual monitoring of the surroundings. Upon intruder detection, the servo motor stops, allowing the camera to focus on the detected area. The farmer can access the live camera feed through the ESP32-CAM IP address to verify whether the intruder is harmful or not.

The farmer can also remotely control the system through the Blynk application, including disabling the fencing circuit, buzzer, and LED indicators when required.

Features

- Automated intruder detection
- IoT-based remote monitoring
- Real-time mobile notifications
- Live camera surveillance
- Remote control through Blynk app
- Automated fencing activation
- Visual and audible intrusion alerts

Outcome

Successfully designed and implemented an IoT-based farm security system capable of detecting intruders, providing real-time alerts, enabling live surveillance, and allowing remote control for improved farm protection and monitoring.